SHENZHEN INSTITUTE
of ARTIFICIAL INTELLIGENCE AND ROBOTICS for SOCIETY
深圳市人工智能与机器人研究院

Recommendation Letter - AIRS Mentor

Date: 26 September 2026

Dear Members of the Admissions Committee,

I am writing to recommend Yongyi Lai for admission to the MSc in Artificial Intelligence and Robotics at The Chinese University of Hong Kong, Shenzhen.

I am a Robot Development Engineer at the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS). I directly supervised Yongyi as a robotics software intern from September 2025 to March 2026, approximately six months. During this period she was responsible for chassis balance control of a three-omni-wheel ballbot carrying a dual-arm upper body, a demanding control problem in our group.

The task was to keep a dynamically unstable base balanced while two arms moved on top of it, continuously shifting the centre of mass. Yongyi built the control state from IMU and wheel feedback, designed a cascaded controller in which a position outer loop generates bounded attitude references for an attitude inner loop, allocated the resulting actions to the three wheels through the wheel kinematics, and deployed and tuned the loop on the physical robot. To handle arm disturbance, she estimated the centre-of-mass offset from arm joint states using the robot's URDF model with Pinocchio and fed it into the balance controller as gravity feedforward. She verified the effect by comparing the balance response with compensation switched on and off under fixed arm trajectories. By the end of the internship, the chassis maintained dynamic balance while both arms executed their trajectories, meeting the objective we had set.

What stood out was Yongyi's ability to return to model-based analysis when empirical tuning could no longer fully explain the robot's behaviour. Rather than continuing to adjust gains without a model, she derived a four-state inverted-pendulum model of the ballbot, linearised it and used LQR to analyse how the weights on position, velocity, tilt and tilt rate trade stability against control effort. That analysis became our reference for interpreting the deployed controller's gains, identifying state anomalies and locating actuator saturation. It demonstrated analytical ability beyond what I would normally expect from an undergraduate intern.

Yongyi was equally reliable on the engineering safeguards that make a hardware control loop trustworthy: ROS state processing and CAN motor execution, state initialisation on re-enable, output limiting, and fall detection and recovery. She implemented these safeguards carefully and documented them clearly, allowing the robot to be tested repeatedly under controlled conditions. She communicated progress and problems concisely, asked precise questions, and worked independently once a direction was agreed.

In my comparative assessment, Yongyi ranked first among the nine undergraduate interns I directly supervised over the past several years in both technical knowledge and suitability for postgraduate study in AI and robotics (1st of 9 in each category; top 11.1%).

Yongyi discussed with me her plan to pursue a taught master's in robotics or AI before working as an algorithm engineer. Given how much of her progress during this internship came from turning to model-based analysis when tuning ran out, I believe this is the right next step. On the evidence of her work, she has the technical ability, hands-on discipline and intellectual curiosity to benefit fully from your programme. I recommend Yongyi without reservation and would be glad to answer any questions.

Yours sincerely,

Referee signature: _____

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